MIL-C- 81317(WP)
30 August 1965

Superseding WS 4374
31 July 1964

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 97 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 97 MOD 0 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-749	Primer, Percussion for Small Arms Ammunition

STANDARDS

Military

MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Bureau of Naval Weapons

DL 2518250	Cartridge, Impulse, MARK 97 MOD 0

Ordnance Corps of the Army

B7645339	Primer, (Styphnate) for Caliber 50

PUBLICATIONS

SECNAV Instruction 4030.3B	Uniform Preservation, Packaging, Packing and Marking of Items of Military Supply

Bureau of Naval Weapons

OP 5	Ammunition Ashore (This publication is for the exclusive use of the Government inspector. It is not to be furnished to bidders or contractors.)
NAVORD Instruction 8010.24	Marking of Federal Stock Numbers and Nomenclature on Ammunition

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite the latest edition and supplements thereto.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 97 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL 2518250 and all documents listed thereon.

3.2 Material. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective BUWEPS drawings unless specific approval in writing covering a departure therefrom has been obtained from the Bureau of Naval Weapons prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 76 MARK 97 MOD 0 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the primer and the main propellant charge are defined as primary components.

3.4 Primer. The primer shall be furnished from lots manufactured within a year past. Primers shall be in accordance with Army Ordnance Dwg. B7645339 and shall be from lots that have passed the tests of MIL-P-749.

3.4.1 X-ray. All primers shall be X-rayed and screened for defects as outlined in 4.7.1.

3.5 Propellant charges.

3.5.1 Weighing. All propellant charges shall be weighed.

3.5.2 Smokeless powder. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. Inspect for signs of decomposition as outlined in paragraphs 1308-1318 and 1322 of OP 5. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

3.5.3 Change of propellant lot. Whenever a change occurs in the propellant lot, validity of the charge determination shall be verified by performing the test prescribed in 4.7.7.

3.6 Cartridge inspection.

3.6.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.7.2.

3.6.2 Leakage. There shall be no leakage of air in the completed cartridge, see 4.7.3.

3.6.3 X-ray. All cartridges shall pass X-ray examination of 4.7.4.

3.7 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.5.1, shall meet the requirements listed below when subjected to the test prescribed in 4.7.5.1.

3.7.1 Maximum chamber pressure. The lower and upper limits of maximum pressure shall be L = 4200 psi and U = 5200 psi, respectively.

3.7.2 Misfire. There shall be no misfires, see 4.7.5.2

3.7.3 Hangfire. There shall be no hangfires, see 4.7.5.3.

3.8 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 76 cartridges shall be prepared. Sixty-six of these cartridges shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for the environmental and functional tests listed below shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.6.3. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in Table III of MIL-D-21625: 40 foot drop (6); 15 g shock (12); salt spray (6); temperature and humidity cycling (18); vibration (12); +70°F (12). No cartridge firing shall produce results outside the limits set forth in paragraphs 3.7.1, 3.7.2 and 3.7.3 of this specification. Firings shall be conducted in the test fixture of 4.7.5.1. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 181 minimum and 500 maximum cartridges. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Production and acceptance inspection. Production and acceptance inspection shall consist of the following:

Item	Nature of Test	Requirement	Test
Primer	X-ray	3.4.1	4.7.1
Charge	Weighing	3.5.1	---
Smokeless Powder	Condition	3.5.2	---
Cartridge	Visual Inspection	3.6.1	4.7.2
Cartridge	Leakage	3.6.2	4.7.3
Cartridge	X-ray	3.6.3	4.7.4
Cartridge	Ballistics	3.7	4.7.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

4.6 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cartridges and their markings conform to Section 5 of this specification.

4.7 Tests.

4.7.1 Primer X-ray. Primers should be mounted in suitable grooved racks. For the most revealing exposure the primers are laid on their side with the anvil cutouts vertical. Pressure sensitive tape may be used as an aid in positioning the primers. Reject any that are questionable from any standpoint, especially for: missing or abnormally thin charge; charge broken, cracked or porous; deformed, misplaced or cracked anvil; or other inert parts.

4.7.2 Visual inspection. Examine each cartridge produced under a contract and reject for dents, deep scratches, split or cracked lip in the crimped area, excess solder, or other defects which may conceivably prevent entry into the firing chamber or affect stability or resistance to moisture.

4.7.3 Leakage. Each cartridge in the ballistic sample or pre-production sample will be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec air at a pressure differential of one atmosphere shall be considered defective. If one or more cartridges of the ballistic sample leaks, then a leak test of each cartridge purchased under contract for the lot being tested shall be required. If a 100% leak test of the lot is required it shall be at the expense of the contractor. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.4 X-ray examination. Examine all cartridges produced under a contract or purchase order by X-ray and any observable imperfections in assembly shall be cause for rejection. X-ray plates may be destroyed after ballistic acceptance of cartridges.

4.7.5 Ballistic tests.

4.7.5.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV and an AQL of 1.0. The cartridges shall be conditioned for 24 hours at +70°F ± 5°F and fired in a closed chamber having an internal volume, with expended cartridge in place of 4.5 cubic inches and adapted for the measurement of pressure. A record of pressure versus time and of elapsed time between actuation of the firing mechanism and start of pressure rise shall be taken by suitable instrumentation for each firing. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

production lot for investigative purposes.

4.7.5.1.1 Criteria of acceptability. The lot shall be acceptable if the requirements of 3.7.1 are met at an AQL of 1.0 under the provisions of Section B, Part II of MIL-STD-414.

4.7.5.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.5.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and start of pressure rise exceeds 0.05 second.

4.7.6 Retest. There shall be no retests.

4.7.6.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and the test reconducted with a new sample of cartridges.

4.7.7 Change in propellant lot. Whenever a change in propellant lot occurs, validity of the charge determination shall be verified by firing 12 cartridges each at temperatures of -65°F, +70°F and +160°F ±5°F. Results of these firings shall fall within the limits prescribed in 4.3.2. If the cartridges fired at 70°F meet the requirements of 4.7.5.1.1 they may be counted as part of the sample required by 4.7.5.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. In addition to the following, Level A and Level C packaging shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. For a definition of Level A packaging, see 6.4.1.

5.1.1.1 Inner container. Twenty cartridges shall be packaged in accordance with Bureau of Naval Weapons drawing 2518260.

5.1.2 Level C. Minimum packaging for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packaging, see 6.4.1.1.

5.2 Packing. In addition to the following, Level A and Level C

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. For a definition of Level A packing see 6.4.2.

5.2.1.1 Outer container. Eight inner containers as described in 5.1.1.1 shall be packed in accordance with the Bureau of Naval Weapons drawing 2518255.

5.2.2 Level C. Minimum packing for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packing see 6.4.2.1.

5.3 Marking. Cartridge marking shall be as specified on Bureau of Naval Weapons drawing 2518250 and NAVORD Instruction 8010.24. In addition, packaging and packing markings shall be in accordance with the explosive section of the Code of Federal Regulations 49 CFR 71-90, MIL-STD-129 and NAVORD Instruction 8010.24.

6. NOTES

6.1 Intended use. The MARK 97 MOD 0 impulse cartridge provides the power to operate the BUWEPS cutters MARK 12 MOD 1, MARK 12 MOD 1 (NM), MARK 12 MOD 2 (NM) and MARK 17 MOD 0.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and if so specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Bureau of Naval Weapons and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from SECNAV Instruction 4030.3B.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use or controlled humidity storage. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations, and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 General safety precautions.

6.5.1 The loading, assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics,etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items shall observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

or safety during use or loading of the specified item, are included in the detail provisions of the specifications.

6.5.2 All loading operations shall be conducted in a neat and orderly manner.

6.5.3 Safe equipment and methods shall be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.4 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning shall insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.5 The exposure of explosive materials and related parts shall be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.6 All explosives and completely or partially loaded items shall be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosive Safety Board covering quantity-distance regulations for explosives, shall apply.

6.5.7 Proper care shall be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.5.7.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.7.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.7.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles shall be conductive to ground electrostatic charges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

6.5.7.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items shall be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (see para. 6.5.5) of work rooms shall be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6.5.7.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric types of primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices shall be employed during manufacturing operations involving electric primers, detonators or squibs.

6.5.7.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.7.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.7.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also clean up any spilled material immediately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81317(WP) CARTRIDGE, IMPULSE, MARK 97 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

BUREAU OF NAVAL WEAPONS (RREN-5)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:11Z
Check the source to verify that this is the current version before use.